MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS

MATH 1101- PRECALCULUS

End of Semester Examination

Date: Wednesday, 14th February, 2018 Time Allowed: 3 hours (08:00-11:00 AM)

INSTRUCTIONS

- Write your registration number, programme and lecturer's name in the spaces provided below
- Check that the paper has **12** questions
- Attempt **all** questions
- Write your answer in the space provided under each question
- Show all your working
- Calculators are not allowed

Registration Number: _____

Programme: _____

Lecturer's Name : _____

Question:	1	2	3	4	5	6	7	8	9	10	11	12	Total
Points:	12	9	9	7	11	11	6	10	4	7	7	7	100
Score:													

1. (a) Factorise $x(2x-1)^{-\frac{1}{2}} + (2x-1)^{\frac{3}{2}}$. [5]

(b) Find all real value solutions for the equation, $9x^{\frac{4}{3}} - 10x^{\frac{2}{3}} + 1 = 0$. [7]

2. (a) Evaluate $\sqrt[4]{\frac{64a^{\frac{2}{3}}}{a^{\frac{1}{3}}}}$. [4]

(b) Rationalise the denominator in $\frac{1}{\sqrt[3]{a}-1}$. [5]

3. Solve the inequality $\dfrac{x+1}{x+2} > \dfrac{x-3}{x+4}$. [9]

4. Determine the domain and range of the function $g(x) = \dfrac{4x-20}{3x-18}$ and present your answer in interval notation. [7]

5. (a) Solve for x, $7^{-4x} = 2^{1+3x}$. [6]

(b) Sketch the graph of the function, $y = |4 - x|$ without using table of values. [5]

6. Evaluate each expression.

(a) $\csc \dfrac{4\pi}{3}$ [5]

(b) $\sin 105°$. [6]

7. Prove the equation $(1 - \sin\theta)(\sec\theta + \tan\theta) = \cos\theta$. [6]

8. (a) Let $g(x) = 2x + 1$. Find $f(x)$, given that $g(f(x)) = 10x - 7$. [5]

(b) Let $f(x) = 2^x$. Find $\dfrac{f(x+h) - f(x)}{h}$ and show that it reduces to $2^x\left(\dfrac{2^h - 1}{h}\right)$. [5]

9. Let $\log_b 2 = A, \log_b 3 = B$ and $\log_b 5 = C$. Evaluate $\log_b 0.6$ in terms of A, B and or C. [4]

10. Suppose that β is an acute angle and $\sin\beta = \dfrac{m^2 - n^2}{m^2 + n^2}$ $(m > n > 0)$. Show that $\cos\beta = \dfrac{2mn}{m^2 + n^2}$. [7]

11. Sketch the graph $f(x) = -cos\left(x + \frac{\pi}{2}\right), \ -2\pi < x < 2\pi$. [7]

12. Solve the equation, $2\sin^2 x = 3\sin x - 1, 0 \leq x \leq 2\pi$ [7]

END OF QUESTIONS